

* Functions:- Set of Instruction or set of code is already predefined is called function.
There are 3 Main Components of function.
They are : (1) Function name.

(2) Number of inputs.

(3) Return value.

(*) FIND Function:- Locates a specific substring within a text string. It is case-sensitive and does not support wildcards.

Syntax: =FIND (find-text, within-text, [start-num])

Example: =FIND("o", "rochitha")

Output: 2 [It finds the lowercase "o" at the second character position].

(*) SEARCH Function:- Similar to FIND, but it is case-insensitive and supports wildcards (like ? and *).

Syntax: =SEARCH (find-text, within-text, [start-num])

Example: =SEARCH ("O", "rochitha")

Output: 2 (It finds "o" despite the uppercase query because SEARCH ignores case)

(*) Split First Name and Last Name:- To split a full name (e.g., "Rochitha Shetty") residing in cell A1 into separate columns.

(i) First Name Formula:

=LEFT(A1, FIND(" ", A1) - 1)

Meaning- Finds the space character, subtracts 1, and extracts everything to the left.

Output: Rochitha

(ii) Last Name Formula:

$=MID(A1, FIND(" ", A1) + 1, LEN(A1))$

Meaning: Starts right after the space character & extracts the remaining text to the end.

Input: In mo

Output: Shetty.

(*) CONCAT Function:- Combines the text from multiple ranges and/or strings (the modern successor to CONCATENATE).

Syntax: $=CONCAT(\text{text1}, [\text{text2}], \dots)$

Example: $=CONCAT("Hi", " ", "Sunshine")$

Output: Hi Sunshine.

(*) TEXTJOIN Function:- Combines text from multiple ranges using a specified delimiter & gives you the option to ignore empty cells.

Syntax: $=TEXTJOIN(\text{delimiter}, \text{ignore-empty}, \text{text1}, [\text{text2}], \dots)$

Example: $=TEXTJOIN(",", TRUE, "red", "green", "blue")$

Output: red, green, blue.

(*) SUBSTITUTE Function:- Replaces specific existing text with new text in a string by matching the exact text characters.

Syntax: $=SUBSTITUTE(\text{text}, \text{old-text}, \text{new-text}, [\text{instance-num}])$

Example: $=SUBSTITUTE("I love cats", "cats", "dogs")$

Output: I love dogs.

Example: $=SUBSTITUTE("Sanjana", "a", "@", [5])$

Output: Sanj@na.

(*) REPLACE Function :- Replaces a portion of a text string with a new string based on a character position and length, rather than matching specific text.

Syntax: =REPLACE(text, start_num, num_chars, new_text).

Example: =REPLACE("PEOPLE", 4, 1, "Q")

Result: PEQPLE.

(*) Transpose :- Flips the orientation of a range of data, converting rows into columns & columns into rows. It can be used as a worksheet function or through the Paste Special menu.

Syntax: =TRANSPOSE(array)

(or) Copy the entire column / row and click Ctrl + Alt + V and select transpose option in that Paste special menu.

Example:

	A
1	Names
2	Rochitha
3	Harshitha
4	Kiran

Copy this entire column from A1 to A4

then select a new cell

& click Ctrl + Alt + V.

then click on Transpose option & click OK.

Output:	B	C	D	E
1	Names	Rochitha	Harshitha	Kiran

(*) Paste Link :- Creates a dynamic formula link between the destination cell & the source cell. If the source data changes, the pasted cell updates automatically.

Syntax : It is a User Interface feature (found under Paste Special > Paste link).

And its underlying formula structure uses an equals sign & the cell structure cell reference.

Example : =Sheet1!A1

This displays the exact value of cell A1 from Sheet1 and updates whenever Sheet1!A1 changes.

- (*) Paste Special Operations (Add, Subtract, Multiply, Divide) :- Allows to perform a mathematical operation on a range of target cells using a copied value all at once, without manually editing each formula.

Syntax : Accessed via the Excel UI (Right click > Paste Special > choose an operation under the "Operation" section like Add, Multiply, Subtract, divide).

Out Example : If you copy a cell containing 10, select a range of numbers, right click, choose Paste Special & select Multiply, Excel will multiply every selected cell's existing value by 10 instantly.

- (*) Absolute References :- Remains constant and locked even when the formula is copied to a different cell, designated by dollar signs.

Syntax : \$column\$Row

Example : \$A\$1.

- (*) Relative References :- Changes automatically when a formula is copied to another cell.

Syntax : ColumnRow.

Example : A1.

= A1*\$B\$1, When you drag this formula down, A1 changes to A2, A3, etc., but \$B\$1 stays fixed as \$B\$1.

(*) Mixed Reference :- Either the column or row is fixed, while the other can change.

There are 2 types :

1. Fixed Column $\Rightarrow$ \$A1.

Column A $\rightarrow$ fixed

Row 1 $\rightarrow$ changes when copied.

2. Fixed Row $\Rightarrow$ A\$1.

Column A $\rightarrow$ changes when copied.

Row 1 $\rightarrow$ fixed.

Example : = \$A8 * G\$2.

This is a mixed reference.

- \$A8 $\rightarrow$ Column A is fixed, row can change.

- G\$2 $\rightarrow$ row 2 is fixed, column can change.

(*) REPT Function :- The REPT function repeats a text or character a specified number of times.

Syntax : = REPT (text, number-times).

Example : = REPT ("*", 4)

Result : ****

(*) SQRT Function :- The SQRT function returns the square root of a number.

Syntax : = SQRT (number)

Example : = SQRT(16)

Output : 4.

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(*) ABS Function:- The ABS function returns the absolute value of a number, meaning it removes the negative sign. Absolute Value (ABS).
Syntax: =ABS (number).
Example: =ABS (-28).
Output: 28.

(*) ROMAN Function:- It converts an Arabic number (ordinary number) into a Roman numeral.
Syntax: =ROMAN (number, [form]).
• number → The number which I want to convert.
• [form] → optional. determines the style of the Roman numeral. If omitted, Excel uses the standard form.
Examples: =ROMAN(5), =ROMAN(10)
Outputs: V, X

(*) ARABIC Function:- The ARABIC function converts a ROMAN numeral into an Arabic number (ordinary numerical value).
Syntax: =ARABIC (text)
• text → The Roman numeral you want to convert. It can be entered directly in quotation marks or referenced from a cell.
Examples: =ARABIC(V), =ARABIC(X)
Outputs: 5, 10.